

Nidhi Agarwal is a Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards and IoT cloud solutions.

MPPT Solar Charge Controller



The reference design optimises solar energy with MPPT and supports multiple lead-acid battery types. It includes load control, safety features, and integrated drivers, making it suitable for solar lighting and off-grid systems.

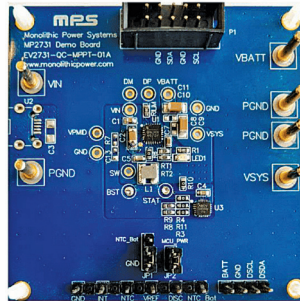
Cypress's MPPT solar charger and LED driver reference design use the PowerPSoC solution to charge solar-powered lead-acid batteries with a maximum power point tracking (MPPT) algorithm that optimises energy harvesting and charging efficiency. This solution also drives LEDs, making it suitable for street and solar lighting applications, offering performance for outdoor and off-grid systems. The MPPT algorithm controls the charging current, allowing the system to operate at maximum power even at night, improving battery efficiency. The design supports various lead-acid battery types, including flooded, absorbed glass mats (AGM), sealed, and gel batteries. The system includes load control options such as automatic dusk-to-dawn operation, battery charging, load enabling, and safety features for the charging circuit, including protection against over-voltage, no battery connection, under-voltage, and low panel voltage. The MPPT algorithm optimises solar energy by modulating charging current to extract power, supporting a 12V lead-acid battery with up to 9.5A charging current. With integrated high-power peripherals, the system includes a boost driver and a buck driver, providing energy management with battery and solar panel protections.

The reference design is suitable for street and solar lighting applications.

Cypress

<https://www.electronicsforu.com/electronics-projects/mppt-solar-charge-controller-reference-design>

Solar MPPT Charger



The solar charger design offers an efficient, small, cost-effective solution with high performance, supporting various input voltages and charge currents. It includes safety features, low power consumption, and flexible configuration options.

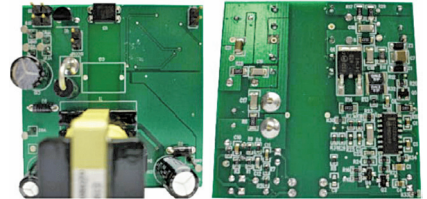
The MP2731 is a solar charger designed for one-cell batteries from Monolithic Power Solutions that helps engineers create small, cost-effective solar charging solutions with high performance. The design includes essential parts like a switcher, sensor circuits, and a power-controlled controller. It uses a unique algorithm to get the most energy from the solar panel, even when the light changes, achieving over 98% accuracy. The charger also has safety features like battery protection and timers. Its small size (25mm x 25mm) fits tight spaces, and engineers can choose parts for the best balance of cost, size, and efficiency. The design works with input voltages from 3.7V to 16V and can handle up to 22V, with up to 4.5A charge current and 93% efficiency. It has low power consumption and an I2C interface to monitor and configure the system.

The reference design is suitable for outdoor solar surveillance cameras and lighting applications.

Monolithic Power Systems (MPS)

<https://www.electronicsforu.com/electronics-projects/solar-mppt-charger-reference-design>

Solar Power MPPT Controller



The reference design simplifies off-grid solar streetlamp design with flexible battery configurations, efficient energy use, and maximum power tracking. It supports solar panels from 10W to 30W, ensuring scalability for various needs.

The solar power MPPT controller reference design from Onsemi simplifies the design process for off-grid solar streetlamps, providing insights for engineers developing solar-powered lighting solutions. The system includes LEDs, a lead-acid battery, a solar controller, and a solar panel that converts sunlight into electrical energy. Sunlight intensity varies throughout the day due to cloud cover, sun angle, seasonal changes, and reflections, causing a solar panel rated for 30W peak power to produce only 24W at midday and 6W in the evening. The panel's voltage output is adjustable, typically 20V to 48V, depending on the solar cells' configuration. The system consumes less than 2mA of the battery, reducing energy usage, and includes maximum power tracking with an error margin of under 5%, optimising energy efficiency. It can charge up to four batteries in series or parallel, offering flexibility in battery configurations. It is compatible with solar panels from 10W to 30W, allowing engineers to scale the design for different power needs.

The reference design suits off-grid solar street lighting, surveillance cameras, e-bike chargers, and outdoor systems.

Onsemi

<https://www.electronicsforu.com/electronics-projects/reference-design-for-solar-power-mppt-controller>